MILITARY SPECIFICATION

POWER SUPPLY PP-2581(*)/A

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope - The equipment covered by this specification shall supply the guided missile with amplification of audio signal, standby power, firing circuit power, motor drive power, and passage of high pressure gas. The high pressure gas will be at 3000 psi working pressure with peak pressure of 4700 psi at elevated temperature.

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-T-27	Transformers and Inductors (Audio, Power, and Pulse)
MIL-E-4682	Electron Tubes and Transistors, Choice and Application of
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing; Environmental, Aircraft Electronic Equipment
MIL-R-5757	Relay, Electrical (Excluding Thermal) for Electronic and Communication-type Equipments, General Specification for
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-W-16878	Wire, Electrical, Insulated, High-temperature
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-T-18303	Test Procedures; Preproduction and Inspection, For Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment

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MIL-L-22620	Launcher, Guided Missile LAU-7/A
MIL-R-22973	Reliability Index Determination for Avionic Equipment Models, General Specification for
MIL-R-23094	Reliability Assurance for Production Acceptance of Avionic Equipment, General Specification for
MIL-D-70327	Drawings, Engineering and Associated Lists

STANDARDS

Military

MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of

DRAWINGS

Bureau of Naval Weapons

DL-60A89D114	Numerical List of Drawings: Power Supply Guided Missile Launcher, PP-2581/A

2.2 Availability of Documents - When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code 105, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern:

(1) Parts and materials requirements shall conform to specification MIL-E-5400.

(2) Nonrepairable subassemblies, as outlined in specification MIL-E-5400, shall be used when practicable. The general size of the subassembly, and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies shall be reliable. (See 6.4)

(3) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.1.1 Nonstandard Parts and Material Approval - Approval for the use of nonstandard parts and materials shall be obtained as outlined in specification MIL-E-5400.

3.1.2 Electron Devices - Transistors and diodes shall be chosen and applied as outlined in Specification MIL-E-4682.

3.1.3 Component Parts - To ensure the high degree of reliability required for the equipment, each electrical component part shall be tested as specified in 4.4.1.2 for conformance to its requirements. Sampling of lots is not acceptable. This requirement applies to all diodes, transistors, relays, transformers, capacitors, resistors and like items used in the power supply. Acceptance or approval of such items by the procuring activity does not guarantee acceptance of finished equipment.

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3.1.4 Wire - The wire used in this equipment shall conform to the requirements of Specification MIL-W-16878, Type E.

3.1.5 Transformers - The transformers in this equipment shall be manufactured in accordance with Specification MIL-T-27 and shall be tested for Group A and B tests as applicable.

3.2 Design and Construction -

3.2.1 General - The equipment shall conform with all the applicable requirements of Specification MIL-E-5400 for Class 3 equipment and DL60A89D114 for design, construction, performance, and workmanship.

3.2.2 Standard Conditions - The following conditions shall be used to establish normal performance requirements and for making laboratory bench tests on the equipment unless otherwise specified in this specification or on the detail drawings:

Temperature	Room Ambient (25 ± 5°C)
Altitude	Normal Ground
Vibration	None
Humidity	Room Ambient up to 90% Relative
Input Power	115 ± 1.0 V AC and 27.5 ± 0.5 V DC.

3.2.3 Service Conditions - The power supply shall operate satisfactorily, when installed in Launcher LAU-7/A and LAU-7A/A, under any of the environmental service conditions as specified in Specification MIL-E-5400 for Class 3 equipment except as modified herein.

3.2.3.1 Tests of Specification MIL-L-22620 - The equipment shall operate satisfactorily when subjected to the following tests of Specification MIL-L-22620:

Temperature	Humidity
Vibration	Salt Spray
Altitude	Temperature Shock
Shock	

3.2.4 Warm-Up Time - No warm-up time shall be required for the equipment prior to operation.

3.2.5 Total Weight - The total weight of the power supply shall not exceed 14 pounds.

3.2.6 Reliability -

3.2.6.1 Operating Life - The equipment shall be capable of operation as specified for not less than 750 hours of actual or simulated aircraft flight time and shall pass the life test specified herein.

3.2.6.2 Reliability In Mean Time Between Failures (MTBF) - The equipment shall have a minimum of 750 hours of mean (operating) time between failures when tested and accepted as outlined under the requirements of 4.4.3.

3.2.7 Primary Input Power - The equipment shall operate as prescribed for nominal power input of 115 V AC, 400 cps, single-phase, and 28 V DC. It shall operate in conformance with the requirements of this specification, when operating from a power source conforming to MIL-STD-704, Category "B", in all respects, except that normal limits shall be:

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(1) 105 to 122 V AC 380 to 420 cps

(2) 18 to 29 V DC

3.2.7.1 Power Requirements - The maximum continuous power required shall not exceed the specified amounts. The a-c load power factor will be approximately 0.6 lagging.

(1) AC Power 115 V AC 230 VA

(2) DC Power 28 V DC 10 amps

3.2.7.2 Overload Protection - Overload protection shall be provided in the equipment. All parts and circuits of the equipment which deliver operating power shall be capable of withstanding low impedance loading including short circuits for the time periods prescribed under paragraph 3.3, and open circuit operations.

3.2.8 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307.

3.2.9 Interchangeability of Reordered Equipment - For reordered equipment interchangeability shall exist between units and all replaceable assemblies, subassemblies, and parts of a designated model of any previously manufactured equipment, supplied or designated by the procuring activity.

3.2.10 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to interference shall be controlled within the limits of Specification MIL-I-6181.

3.2.11 Connectors, Electrical - The equipment shall provide for the use of connectors in accordance with Specification MIL-F-5400.

3.3 Performance - The unit shall be capable of performing the following functions:

3.3.1 Standby Missile Power - Provide standby electrical power for the missile by utilizing 115 volts, 400 cps, single-phase power defined in 3.2.7 to supply the following:

3.3.1.1 Low-Voltage Supply - DC voltage for audio amplifier and vacuum tube filaments.

(1) The voltage supplied to the missile shall be 25.2 plus or minus 1.2 volts over an output current range from 0.5 to 1.5 amps under any normal steady state condition of MIL-STD-704 input power as defined in 3.2.7.

(2) The open circuit voltage shall not exceed 27.0 volts dc steady state.

(3) These requirements shall apply over the specified temperature range.

(4) The maximum ripple, under standard conditions with 1.5 amps load shall be equal to or less than 0.15 volt peak-to-peak. Under any condition of voltage, frequency, load, and temperature the ripple shall not exceed 1 volt peak-to-peak. All ripple voltages consisting of frequencies other than the fundamental and second harmonic of the input AC supply shall be ignored.

(5) Under any transient of input voltage specified in 4.2.4.2.4, the output voltage shall not exceed 40.5 $\pm$ 0.5 volts dc.

(6) The output voltage with an output current of 3.0 amps (30 sec max.) shall not be less than 21 volts.

(7) The 25.2 V DC power supply shall be capable of withstanding continuous short circuit loading without damage.

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3.3.2 Audio Amplifier - Amplify the audio signal from the missile and feed it to the pilot's headset.

(1) The amplifier shall have a gain of 0.75 to 1.5 at 1500 cps over an input range of 0.5 to 24 volts rms. It shall not saturate at any input up to 24 volts with a 1000 ohm load. The gain shall be within 3db over the frequency range of 400 to 2000 cps with an input voltage of 10 volts rms.

(2) Input impedance shall be a minimum of 20K ohms. Output impedance shall be not greater than 1000 ohms.

(3) Distortion shall not exceed 10 percent for any input from 0.5 to 24 volts over a frequency range of 400 to 2000 cps.

(4) The random noise across the 1000 ohm load shall not exceed 0.15 volt rms with the input shunted by a 20K ohm resistor.

(5) The above requirements shall be maintained over the full operating temperature range.

3.3.3 Series-Tuned Circuit - This circuit shall be constructed in accordance with BuWeps drawing 60A89C217.

(1) Relay K-3 shall pull in when an rms voltage of not greater than 40 volts at a frequency of 5.700 ± 0.05 kc is applied between the input to the circuit and ground.

(2) Relay K-3 shall not pull in when an rms voltage of less than 80 volts at a frequency of 4000 ± 0.05 kc is applied between the input to the circuit and ground.

(3) The voltage across the series-tuned circuit shall be between 155 volts peak ± 5 volts and 205 volts peak ± 5 volts either positive or negative.

(4) All the above shall apply over the specified temperature range.

3.3.4 Gas System - A feedthru gas system shall be provided to allow the passage of dry air or nitrogen for missile cooling. The gas system shall have a maximum leakage rate of 1 cc per minute at a pressure of 4700 (plus or minus 50) psi at 77 (plus or minus 10) degrees F. The pressure drop across the feedthru gas system shall be less than 35 psi at a flow rate of 10 standard liters per minute. The cleanliness of the gas system shall be in accordance with MIL-L-22620.

3.3.5 Schematic Diagram - The power supply shall be wired in accordance with basic schematic diagram BuWeps drawing 60A89F125. Pin assignments shall be as indicated.

3.3.5.1 Feedthru Wiring - Feedthru wiring shall be provided for the following missile functions:

(1) 115 V 400 cps
(2) Missile Identification
(3) TM on-off
(4) Power interlock relay
(5) Acquisition coil No. 2
(6) AIM-9B Acquisition coil No. 1
(7) AIM-9B Spare
(8) AIM-9B G and C
(9) AIM-9B Acq ref r-1
(10) AIM-9B Acq lambda
(11) AIM-9B Acq ref u-d
(12) Ground
(13) N_2 Valve ground

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3.3.5.2 Jettison - A jettison relay (K-1) shall be provided for emergency release of an unarmed and unguided missile or target rocket.

3.3.5.3 Firing and Master Arm Circuit - A firing and master arm circuit, as indicated in BuWeps drawing 60A89F125, shall be provided for firing of an armed and guided missile or target rocket. Safety interlocks indicated are required.

3.3.5.4 Manual Uncage - A manual uncage relay (K-5) shall be provided to open the missile caging loop upon a plus 28 vdc signal from the aircraft.

3.3.5.5 AN/ASA-38(V) Lock-On Relay - A relay (K-6) shall be provided to open the electrical caging lead into the aircraft upon signal from the aircraft.

3.4 Design Data - No data is required by this specification (other than reports accompanying samples submitted for testing) or by applicable documents, unless specified in the contract or order (See paragraph 6.6).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(1) Preproduction Tests
(2) Production Sample Tests
(3) Acceptance Tests
(4) Life Tests

4.2 Preproduction Tests - Preproduction tests shall be made on ten equipments representative of the production equipments to be supplied under the contract. Preproduction tests shall consist of the following:

(1) Contractor Demonstration Tests
(2) Service Approval Tests
(3) Reliability Index Determination Tests.

4.2.1 Contractor Demonstration Tests - Contractor demonstration tests on three of the equipments shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.8. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory.

4.2.1.1 Demonstration Test Data - The contractor shall submit all data collected in conducting these tests to the procuring activity for review and approval. On first production contracts this data shall include a list of all electrical and electronic parts giving their specified voltage, current and temperature rating and the applied circuit voltage, current and ambient and surface temperatures. The ambient and surface temperature shall be obtained under the extreme high temperature operating condition. Such data on parts may be included as part of the preliminary parts list.

4.2.2 Service Approval Tests - At the completion of the contractor demonstration tests, the three equipments shall be delivered at the contractors expense to a specified government laboratory for additional testing. This additional testing may consist of duplicating tests previously conducted and such other tests as are deemed necessary to determine compliance with all applicable design and performance requirements.

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4.2.2.1 Accessory Material - In addition to the complete equipments submitted for service approval tests the contractor shall also submit the accessory material and design and test data specified in Specification MIL-E-5400. This information shall indicate the physical and electrical characteristics of the equipment and establish the equipment's compliance with applicable requirements.

4.2.3 Reliability Index Determination Tests - At the completion of the contractor demonstration tests, the remaining seven preproduction equipments shall be delivered at the contractor's expense to a laboratory to be specified by the government for Reliability Index Determination Tests.

4.2.4 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with the procedures of Specification MIL-T-5422, interference tests in accordance with Specification MIL-I-6181, reliability index determination in accordance with MIL-R-22973, and the exceptions and modifications below.

4.2.4.1 Equipment Stabilization - When equipment stabilization is required as specified in various tests, the following method shall be used to determine the time of stabilization. Resistance measurements shall be taken across J-1 pin 18 ground (missile signal output transformer). Measurements shall be taken every ten minutes until duplicate readings have been obtained. Stabilization has been established when resistance measurements are repetitive.

4.2.4.2 Operating Tests - Operating tests of the power supply only (not installed in a launcher) shall check all functions and shall include (but not be limited to) the tests specified in the following paragraphs. Specific procedures for operating tests shall be prepared by the contractor and shall be subject to approval of the procuring activity (see 4.8).

4.2.4.2.1 Ripple Voltage Measurement - Ripple voltage measurements shall be taken with a maximum of three feet of cable attached to J1.

4.2.4.2.2 Power Input - Table below lists the various power inputs which shall be applied during testing.

	POWER CONDITION:		
Test:	A	B	C
A-C Voltage	115 ac	105 ac	122 ac
Frequency	400 cps	420 cps	380 cps
Low Voltage DC	18 and 29 vdc	18 vdc	18 vdc

4.2.4.2.3 Load Conditions - External load conditions listed in the table below shall be applied during testing:

	LOAD CONDITION:			
Test	A-No Load	B-Min. Load	C-Full	D-Over
175 VDC	0	125 ma	175 ma	280 ma (60 sec max)
25.2 VDC	0	0.5 amp	1.5 amp	3A (30 sec)

4.2.4.2.4 Transient Test - With the power supply operating with input power as defined in 4.2.4.2.2 Condition C and the power supply loaded as defined in 4.2.4.2.3 Condition A the ac input voltage shall be raised from 0 volts to 190 volts rms, maintained for a period of 100 (plus or minus 5) milliseconds and then dropped to test condition value. The rise and fall times shall not exceed 3 cycles. The output voltages shall not exceed allowable maximums. Under the previous conditions of input power and loading the input voltage shall be raised from 120 volts to 190 volts rms, maintained for a period of 100 (plus or minus 5) milliseconds and then dropped to 120 volts rms.

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The rise and fall times shall not exceed 3 cycles. The output voltages shall not exceed allowable maximums. The above cycle shall be repeated three times.

4.2.4.3 Insulation Resistance Test - Each power supply submitted shall be subjected to an insulation resistance test wherein a high potential is applied between each isolated circuit and other circuits, and between each isolated circuit and ground. However, circuits containing diodes or other parts that may be damaged by the high potential are exempt from this test. The following conditions apply:

(1) A high potential of 500 (plus or minus 50) volts dc shall be applied.

(2) The insulation resistance shall be 100 megohms minimum.

4.2.4.4 Gas System - The gas system shall be tested with dry air or nitrogen at a pressure of 4700 (plus or minus 50) lbs std at 77 (plus or minus 10) degrees F for leakage and at a pressure of 3000 (plus or minus 50) lbs standard for flow.

(1) All gas used for the gas system tests shall have a dew point of not more than minus 40 degrees F, shall have a total hydrocarbon content of not more than 50 ppm by volume, and shall contain no particles greater in size than 3 micron nominal and 15 micron absolute.

(2) With a suitable pressure system at a pressure of 4700 (plus or minus 50) lbs standard, a leakage test shall be performed and the maximum leakage rate shall not exceed 1 cc per minute.

(3) With a suitable rotometer and pressure monitoring system a pressure drop shall be performed. The total pressure drop across the feedthru gas system shall be less than 35 psi standard at a flow rate of 10 standard liters per minute.

4.2.4.5 Operational Tests - Testing of the power supply (with the power supply installed in the launcher) shall be in accordance with 4.2.4.2. Power inputs specified in 4.2.4.2.2 and load conditions specified in 4.2.4.2.3 apply. Tests shall be conducted with all loads and power inputs.

4.2.4.6 Functional Tests - Testing of the power supply (with the power supply installed in the launcher) shall be in accordance with 4.2.4.2. Power input specified in 4.2.4.2.2 Condition A and load specified in 4.2.4.2.3 Condition C apply.

4.2.4.7 Radio Noise Interference - The power supply shall be tested for conformance to Specification MIL-I-6181 as follows:

(1) The power supply shall be installed in the launcher and tested for radio noise interference.

(2) A set of data shall be obtained for condition C of 4.2.4.2.3.

(3) A dummy missile with standard umbilical shall be used during these tests. This missile shall be provided with a special harness allowing connection of test equipment during these tests. The maximum test equipment cable length shall be nine inches.

4.2.4.8 Environmental Tests - Environmental tests shall be performed with the power supply installed in the launcher and shall be conducted as specified in Specification MIL-L-22620. Prior to conducting any environmental tests, the equipment shall be subjected to the operational tests of 4.2.4.5.

4.2.4.9 Preproduction Life Test - The equipment shall be subjected to a life test under the following conditions: The equipment shall be mounted in a LAU-7/A launcher or equivalent heat sink during this test. The operating time in the preceding tests and the 200-hour life test shall total not less than 500 hours, altitude shall be normal ground. During 200-hour life tests, or until completion of 500 hours, the input power shall be standard as defined in 3.2.2. The equipment shall

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be placed in a temperature chamber at ambient temperature. With the equipment operating, the chamber temperature shall be raised to 130 degrees F, stabilized within a period of one hour and maintained for a period equal to four hours minus the stabilizing time. The temperature of the chamber shall then be lowered to minus 65 degrees F, stabilized within a period of one and one-half hours and maintained for a period of four hours minus the stabilization time. This temperature cycle shall be repeated for a period of 200 hours and tested with the loads as indicated in 4.2.4.2.3.

At the completion of the above testing, if the total time is less than 500 hours as specified, the unit shall be operated continuously at 4.2.4.2.3 Condition C loads at room ambient to complete the 500 hours.

4.2.4.9.1 Test Period - The test may be run continuously or intermittently. However, not less than one full cycle of 8 hours shall be conducted at one time. Simulated missile firings shall be continued to a minimum of 1700 times. The power ON-OFF and firing cycle shall be as follows:

(1) Power on for 5 minutes

(2) Fire (do not hold firing power on for more than 5 seconds)

(3) Power off for 2 minutes

Repeat this cycle continuously until the specified time and firings have been completed.

4.2.4.9.2 Performance Check - A performance check shall be made during the first temperature cycle, and every third cycle thereafter. An operational test shall be made in accordance with 4.2.4.5. This test shall be conducted alternately during the last part of the high and low temperature portions of the cycle.

4.2.5 Preproduction Approval - Approval of the preproduction samples shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction samples. Prefabrication of production equipment prior to the approval of the preproduction samples is at the contractor's own risk. The approved preproduction samples will be returned to the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction samples shall not be considered as equipments under the contract.

4.2.6 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction samples. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction samples. Evidence of non-compliance with the above shall constitute cause for rejection and for equipment already accepted by the government it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Production Sample Tests - When requested by the procuring activity, one equipment shall be selected from the first 10 production equipments submitted for acceptance and sent at the contractors expense to a designated government laboratory for tests. This equipment shall be selected by the government inspector after the equipment has successfully passed all individual tests. The preproduction sample shall not be selected for this test.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction samples in design, construction, workmanship, performance and quality and that it meets all applicable requirements.

4.3.2 Production Sample Approval - Approval of the production sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the production sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

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4.3.3 Reconditioning of Production Test Sample - On completion of the production sample test the equipment shall be reworked by the contractor by replacing all worn or damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing all acceptance tests except the reliability assurance tests. All inspection and testing shall be under the supervision of the Government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the finished product. Acceptance tests shall consist of the following:

(1) Preliminary assembly and component parts tests

(2) Individual tests

(3) Reliability Assurance Tests

(4) Special Tests

4.4.1 Preliminary Assembly and Component Tests - Tests shall be performed on each component of the power supply prior to assembly in accordance with the following.

4.4.1.1 Preliminary Assembly Tests - The equipment shall be assembled from satisfactory components and prior to potting shall be tested in accordance with 4.2.4.3.

4.4.1.2 Component Tests -

4.4.1.2.1 Transformer and Choke Tests - Each transformer and choke shall be tested to determine whether resistance, inductance and winding ratios, where applicable, are within the values given on the detailed drawing. High potential break-down tests shall be performed on each item as listed on the detailed drawing to determine adequacy of insulation between windings and/or core as applicable.

4.4.1.2.2 Capacitor and Resistor Test - Tests shall be performed on each capacitor and resistor to determine whether capacitance and resistance values are within the values given on detailed drawings.

4.4.1.2.3 Transistor and Diode Test - Each transistor and diode shall be tested at room temperature to determine conformance with the typical operating curves published by the manufacturer of the device. Each device shall be temperature cycled ten times and retested to determine that no electrical deterioration has occurred. A cycle is defined as follows: The diode case temperature shall be changed from room ambient temperature to minus 65 degrees F in one minute maximum, from minus 65 degrees F to 212 degrees F in two minutes maximum, and from 212 degrees F to room ambient temperature in one minute maximum. The complete cycle shall be accomplished in four minutes.

4.4.1.2.4 Relay Test - Each relay used in the power supply shall be subjected to the following tests. The tests shall be performed in the sequence specified.

4.4.1.2.4.1 Contact Test - A test shall be made to determine whether the relay contacts transfer or chatter during vibration in each of three planes at an amplitude of 0.125 inch (double amplitude of 0.250 inch) over a frequency range of 10 to 28 cps and at an acceleration of 10g over a frequency range from 28 to 2000 cps in no less than ten minutes per plane. The test shall be conducted with the relays in the energized, and then de-energized condition; 18 vdc maximum shall be used to energize the relays.

4.4.1.2.4.2 Pick-Up and Drop-Out Voltage Test - Pick-up and drop-out voltages shall be as specified on drawings after the relay has been stabilized at temperatures of minus 65 degrees F, 77 degrees F, and 230 degrees F.

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4.4.1.2.4.3 Dielectric Test - A dielectric strength test shall be conducted in accordance with Specification MIL-R-5757.

4.4.2 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment shall have passed the following tests:

(1) Examination of product

(2) Insulation resistance test

(3) Gas pressure system test

(4) Operating test

(5) Manufacturing Run In test

4.4.2.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.2.2 Insulation Resistance Test - An insulation resistance test shall be performed in accordance with 4.2.4.3.

4.4.2.3 Gas Pressure System test - The gas pressure system shall be tested in accordance with 4.2.4.4.

4.4.2.4 Operating Test - Each power supply shall be subjected to an operating test in conformance with 4.2.4.2.

4.4.2.5 Manufacturing Run In Test - Each equipment shall be operated under the conditions specified herein for a period of 6 hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

Temperature	Ambient room
Humidity	Ambient room
Vibration	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ± 3 g's
Power and Loads	Condition A of 4.2.4.2.2 and C of 4.2.4.2.3

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to that plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions the vertical direction shall be used. During the 6-hour period of operation following the 10-minute vibration period simulated missile firings shall be conducted at 20 minute intervals. Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from all undelivered equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

4.4.3 Reliability Assurance Tests - Reliability assurance tests shall be conducted as required by Specification MIL-R-23094, at a test facility designated by the procuring activity.

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Equipments submitted for reliability assurance tests shall first have passed the individual tests.

4.4.3.1 Reliability Procedure - Reliability Procedure I from Specification MIL-R-23094 shall be used.

4.4.3.2 Test Level - The Test Level shall be Level III from Table I of Specification MIL-R-23094. During each cycle the equipment shall be operated in accordance with 4.2.4.2 during the heating period of each cycle.

4.4.3.3 Accept-Reject Criteria - Figure 15 of Specification MIL-R-23094 shall be used to determine the accept-reject criteria for the first production lot, and Figure 16 shall be used for subsequent lots.

4.4.3.4 Length of Heat Portion of Cycle - After stabilization at the high temperature limit required by the test level, the equipment shall be operated during each cycle as necessary to assure that each equipment will undergo a complete operating test of 4.2.4.2 at least once during each 24 hours of the test.

4.4.3.5 Performance Characteristics to be Measured

Characteristics	Spec Requirement	Acceptance Requirement (if lower)	Time in cycle that measurement is to be made
Operating	4.2.4.2	No degraded performance acceptable	During Period A starting at lowest test level temperature and also during stabilized high temperature, Period B. Refer Table I of MIL-R-23094

4.4.3.6 Failure Criteria - In addition to the requirements of Specification MIL-R-23094, the following requirements shall be used to determine when a failure has occurred during the test:

(1) Whenever performance characteristics fall below the Acceptance requirement (paragraph 4.4.3.5) at least one failure has occurred. If subsequent analysis reveals that several parts have deteriorated, each shall be counted a failure, unless one caused the other parts to fail.

4.4.3.7 Preventive Maintenance - There are no preventive maintenance measures applicable to the equipment.

4.4.4 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect on any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the reliability assurance tests.

4.4.4.1 Special Test Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early equipment after an engineering or material change.

(2) Whenever failure reports or other information indicate additional tests are required. (This will be determined by the procuring activity).

4.4.4.2 Scope of Tests - Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.4.5 Equipment Failure - Should a failure occur during either the reliability assurance or special tests, the following action shall be taken:

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(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Life Test - The contractor shall furnish all samples and shall be responsible for delivery of the life test samples to a Government designated laboratory for tests. The life test sample shall be selected by the Government inspector in accordance with the following; except that when a production sample equipment is requested and submitted to the procuring activity for test then the first equipment specified below need not be selected or subjected to this test. (Equipments which have successfully passed the reliability tests or special tests may be selected for life tests.)

Quantity of Equipments Offered for Acceptance	Quantity to be Selected For Life Test
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

4.5.1 Test Conditions - The life test shall be conducted in accordance with 4.2.4.8

4.5.2 Test Data - A daily record shall be kept of the performance of the equipment, making particular note of any deficiencies or failures throughout the test. This record shall indicate the following:

(1) Part type number

(2) Circuit reference symbol number

(3) Part function

(4) Name of manufacturer

(5) Nature of the failure

(6) Number of hours which the part operated prior to failure

4.5.2.1 Failure Report - In the event of a failure the procuring activity shall be notified immediately and provided with detailed information as to the nature of the failure, to serve as a basis for determination of desired action.

4.6 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply to the best of his knowledge and belief with all applicable requirements. The design data submitted by the contractor with the preproduction model shall indicate its physical and electrical characteristics and other necessary test data which will establish that the equipment meets all the requirements of this specification.

4.7 Reconditioning of Sampling and Special Test Samples - Power supplies which have been subjected to the sampling, special, and life tests will be returned to the contractor, upon request, and may be reconditioned as necessary and resubmitted for acceptance. Before resubmitting, full particulars concerning the rework shall be furnished the Government Inspector. Power supplies rejected after retest shall not be resubmitted without the specific approval of the procuring activity.

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4.8 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests, and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the Government Inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When test procedures are available from previous contracts such procedures may be used when approved by the procuring activity. Unless specifically waived by contractual action the detailed procedures specified herein shall be included where applicable.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-E-17555.

6. NOTES

6.1 Use - Power Supply PP-2581/A is for use in Launcher, Guided Missile LAU-7/A which in turn is used to carry, launch, and jettison missiles and target rockets.

6.2 Test Values - Normal and limiting values of performance data shall be determined at input voltages of 27.5 ± 0.5 V DC and 115 ± 1.0 V AC as applicable. These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standard of performance.

6.3 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration.

6.4 As a general rule non-repairable subassemblies should be encapsulated or hermetically-sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions (such as high voltage power supplies), the non-repairable subassembly should evidence a Mean-time-to-failure greater than 5000 hours, and for many applications this figure must be nearer 50,000 hours.

6.5 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.6 Ordering Data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(1) Title, number, and date of this specification.

(2) Selection of applicable levels of packaging and packing (see 5.1).

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(3) Data requirements (see 3.4).

6.7 The parentheses (*), when used in the type designation, will be deleted or replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.2.8. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-P-81001 POWER SUPPLY PP-2581(*)/A

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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Bureau of Naval Weapons
Washington, D. C. 20360

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